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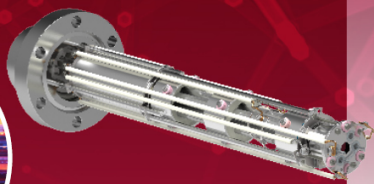
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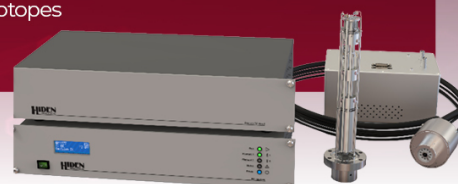
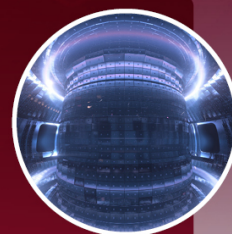
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The influence of Fe-ion irradiation on the microstructure of reduced activation ferritic-martensitic steel Eurofer 97

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Abstract

Reduced activation ferritic/martensitic steels for in-vessel components of a fusion reactor have shown a decrease in plasticity and radiation hardening at low irradiation temperatures. The formation of dislocation loops and embryos of α' phase is considered the main reason for these effects. In this work, Eurofer 97 steel was irradiated with 5.6 MeV Fe^{2+} ions up to 10^{20} m^{-2} at 250, 300 and 400 °C. Transmission electron microscopy study of ion irradiated samples revealed nucleation of dislocation loops. The pair-correlation analysis of atom probe tomography data detected an initial stage of solid solution decomposition. The hardening of ion-irradiated Eurofer 97 was calculated with the dispersed barrier hardening model, taking into account radiation-induced dislocation loops to compare it with the change of yield stress in neutron-irradiated Eurofer 97. According to the obtained results, it can be supposed that the formation of dislocation loops plays the main role in the low-temperature radiation hardening of Eurofer 97 at a dose level up to ~ 10 dpa.

Keywords: radiation hardening, Eurofer 97, dislocation loop, ion irradiation, TEM, APT

(Some figures may appear in colour only in the online journal)

1. Introduction

The reduced activation ferritic/martensitic (RAFM) steels are considered as promising structural materials for advanced fusion reactors working under extreme operational conditions at high neutron fluxes ($\sim 10^{13}$ – $10^{14} \text{ n (cm}^2 \text{ s)}^{-1}$) and at elevated temperatures (up to 650 °C). F/m steels show reasonably good thermo-physical and mechanical properties, a low sensitivity to radiation-induced swelling under neutron irradiation and good compatibility with major cooling and breeding materials [1]. The RAFM steel Eurofer 97 is a European benchmark structural material for in-vessel components of fusion reactors, on which extensive studies have been carried out. When

irradiated to 2.5, 8.4 and 16.3 dpa in the high flux reactor (HFR, Petten) and irradiated to 32 and 70 dpa in the BOR 60 (Dimitrovgrad) reactor at temperatures of ~ 300 °C, Eurofer 97 demonstrated degradation of its mechanical properties, i.e. irradiation-induced strengthening and therefore the loss of ductility [2, 3]. Detailed transmission electron microscopy (TEM) analysis of its microstructure was carried out in [4, 5]. TEM-visible irradiation microstructural features (voids and dislocation loop formation, MX and M_{23}C_6 precipitate evolution, and grain boundary segregation) can only partly explain the hardening of irradiated Eurofer 97 [5, 6]. The atom probe tomography (APT) study of Eurofer 97 irradiated in BOR 60 up to 32 dpa at 330 °C revealed nucleation of numerous nuclei

of α' -phase [7], which can give additional input to radiation-induced strengthening [8].

The objective of the present paper is to provide an analysis of the microscopic origin of degradation of Eurofer 97 under low-temperature irradiation. Reactor experiments on radiation resistance of materials are very expensive and time consuming. Nowadays, simulation experiments with heavy ion beams are providing some progress in the understanding of radiation degradation of f/m steel. Self-ion irradiation was used in this study of radiation effects in Eurofer 97.

2. Materials and methods

An industrial batch of European RAFM steel Eurofer 97 was produced by Böhler Austria GmbH. To obtain a well-defined martensitic–ferritic structure, Eurofer 97 was subjected to a two-step heat treatment at about 980 °C 30 min/air, followed by 760 °C 2 h/air (the heat E83697). This very material was used for the ARBOR and SPICE irradiation programs. The composition of Eurofer 97 steel is presented in table 1.

Disks with a diameter of 3 mm and 0.1 mm thickness were cut out from the bulk material of the Eurofer 97. These disk specimens were mechanically polished up to a thickness of 100 μm . The roughness of the sample surface was measured with atomic force microscopy before and after irradiation, and it was less than 50 nm for all samples.

In our study, Fe^{2+} ion beam accelerated up to 5.6 MeV was used for irradiation [9]. According to the SRIM calculations, the maximum number of generated defects in Fe–9Cr under 5.6 MeV Fe^{2+} irradiation is located at about 1.4–1.5 μm depth (see figure 1). A beam profile monitor was used to analyze the collected ion fluence and homogeneity of ion irradiation [10]. Fluence inhomogeneity on samples did not exceed 15%. To control the specimen temperature continuously during irradiation, two thermocouples were used, which were placed near the specimen. In addition, temperature profiles were calculated to simulate the irradiation conditions. It was shown that the specimen temperature did not vary by more than 5 °C under the irradiation. The disk specimens were irradiated up to 10^{20} m^{-2} at 250, 300 and 400 °C. An estimated dose in the damage peak for this fluence was about 10 dpa.

An analysis of the chemical and phase composition was carried out by TEM, electronic diffraction, high-resolution TEM and scanning transmission electron microscopy (STEM). A Titan 80–300 S/TEM (Thermo Fisher Scientific) microscope with the accelerating voltage of 300 kV and equipped with a high-angle annular dark field detector (Fischione) to provide Z-contrast micrographs was used. The qualitative and quantitative chemical analyses of samples were carried out by energy dispersive x-ray spectroscopy.

Cross-sectional samples for S/TEM studies were prepared by focused ion beam (FIB) on a FIB/scanning dual beam microscope Helios NanoLab 600i (Thermo Fisher Scientific). Samples for APT were also cut and lifted out with FIB from the irradiated samples at various depths from the irradiated surface. APT studies were carried out with an APPLE-3D (ITEP) [11] atom probe microscope in laser evaporation mode

Table 1. Chemical composition (wt.%) of Eurofer 97 steel heat E83697.

C	Si	Mn	Cr	Ni	N	V	W	Ta
0.12	0.06	0.47	8.93	0.022	0.018	0.2	1.07	0.14

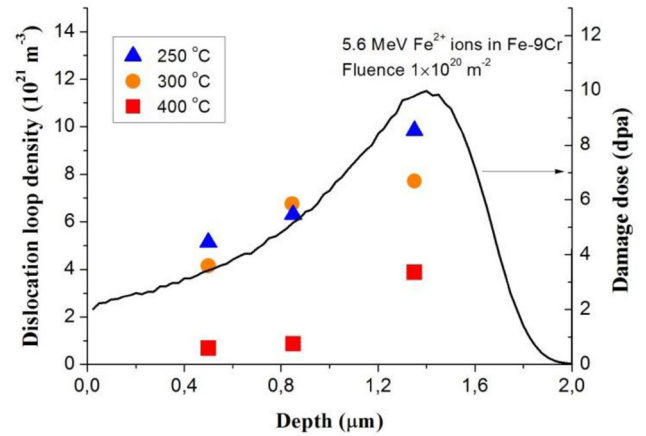


Figure 1. SRIM damage (solid line) profile in Fe–9Cr irradiated with 5.6 MeV Fe^{2+} ions and loop number density (points) at different irradiation temperatures. SRIM calculation was carried out for a dose of $1 \times 10^{20} \text{ m}^{-2}$, using the threshold displacement energy of 40 eV and the Kinchin-Pease option.

with the wave length of 515 nm. Data collection was carried out at the sample temperature of 50 K, laser frequency of 25 kHz and pulse energy of 0.1–5 μJ .

3. TEM/STEM microstructure analysis

The microstructure analysis was performed on both unirradiated specimens and those irradiated with 5.6 MeV Fe^{2+} ions up to 10^{20} m^{-2} at 250, 300 and 400 °C. For the quantification of radiation-induced defects, bright field (BF) and dark field images were acquired.

The microstructure of the Eurofer 97 steel in the as-received condition consisted of tempered martensite with a prior austenite grain size in the range between 6 and 11 μm . A small fraction of spherical particles with the size of 30–90 nm enriched in W and Ta was observed inside the grain structure. In addition, large disk-shaped carbides were detected at the boundaries between the martensite plates. At no irradiation temperatures used in the experiment, were any significant microstructural changes (prior austenite grain size, morphology, size and distribution of the second phase precipitates) observed by TEM. Radiation-induced defect clusters and dislocation loops were observed in all irradiated specimens. In Fe-based ferritic steels, two types of dislocation loops with Burgers vectors $\frac{1}{2}a\langle 111 \rangle$ and $a\langle 100 \rangle$ were found. Specifics of the dislocation loop quantification and microstructure analysis of irradiated Eurofer97 was presented by Dethloff *et al* in [12].

Micrographs for the estimation of the dislocation loop density and diameters were acquired from the grains with orientation close to the $\langle 111 \rangle$ zone axis. In addition, the dislocation loops were highlighted using weak beam dark field (WBDF) imaging with $g(3g)$, $g = 110$ diffraction condition. According

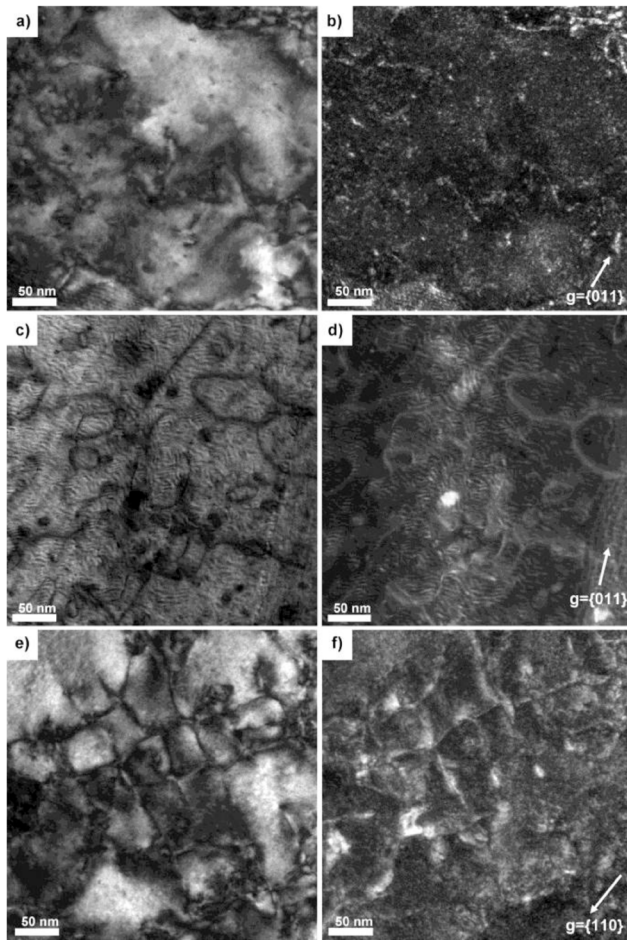


Figure 2. BF (a), (c), (e) and (g, 3g) WBDF (b), (d) and (f) images under [1 1 1] zone axis showing radiation-induced defects in Eurofer 97 irradiated with 5.6 Fe²⁺ ions to 5 dpa at 250, 300 and 400 °C accordingly.

to $g \cdot b = 0$ visibility criteria $2/3$ of $a\langle 100 \rangle$ and $1/2$ of $\frac{1}{2}a\langle 111 \rangle$ loops could be detected under these conditions. Figure 2 shows the microstructure of the Eurofer 97 sample after irradiation with Fe²⁺ ions at different temperatures. On the micrographs, they had a contrast as black dots or typical ‘coffee bean’ pattern that could be highlighted using the WBDF method. The calculated number density of defects at various depths correlated with the radiation damage profile (figure 1). To estimate the thickness of the areas chosen for quantification, the converged beam diffraction technique was used. An analysis of the radiation-induced changes was carried out at the depth of ~ 0.5 , 0.9 and $1.3 \mu\text{m}$, where damage doses were ~ 3 , 5 and 10 dpa, respectively (see table 2). The calculations showed that with the increase in irradiation temperature, the density of dislocation loops reduced. The loop diameter increased with increasing the irradiation temperature at a depth of 0.9 and $1.3 \mu\text{m}$ (table 2).

4. Atom probe tomography

The comparison of the atom maps obtained for un-irradiated and ion-irradiated specimens did not show any signs of visible

Table 2. Number density and average size of dislocation loops at different depths in Eurofer 97 irradiated with 5.6 MeV Fe²⁺ ions to $1 \times 10^{20} \text{ m}^{-2}$.

Depth (μm)		0.5 ± 0.1	0.9 ± 0.1	1.3 ± 0.1
Dose (dpa)		3 ± 1	5 ± 1	10 ± 1
Loop number	250 °C	50	63	98
density (10^{20} m^{-3})	300 °C	40	67	77
	400 °C	7	8	38
Loop average size	250 °C	5	7	6
(nm)	300 °C	9	10	10
	400 °C	20	19	15

precipitation or clustering processes. In some volumes, large features such as grain boundaries or carbides were detected. The homogeneity of the atom spatial distribution was tested using pair correlation function (PCF) analysis [13]. The correlation analysis was completed in $15 \times 15 \times 30 \text{ nm}^3$ volumes extracted away from the large features. The result of APT analysis of samples lifted out at $\sim 1 \mu\text{m}$ depth from the irradiated surface is shown in figure 3. A slight increase ($\sim 5\%$) in the pair Cr–Cr correlation function was observed at small distances ($< 4 \text{ nm}$) in the specimens irradiated at 400 °C . An increase in the Mn–Mn PCF was observed at small distances at all irradiated temperatures, but the biggest value of Mn–Mn correlation was detected at 400 °C . This effect corresponds to the nucleation of chromium-enriched embryos (clusters enriched in Cr and Mn partially). The important role of the correlation between Cr and Mn atoms in the nucleation of α' precipitates was mentioned in the case of neutron-irradiated Eurofer 97 at 330 °C to 32 dpa, where clusters enriched in Cr higher than 20% were detected [7, 8]. However, the detected increase in the PCFs in our study is considerably lower than the one detected in Eurofer 97 irradiated with neutrons at 330 °C to 32 dpa. This allows us to suggest that in the steel irradiated with heavy ions up to ~ 10 dpa, chromium-enriched embryos make only a small contribution to hardening.

To evaluate this input to hardening, the number density of Cr-enriched embryos was estimated by an analytical model of the PCF [13]. According to the behavior of $G_{\text{Cr-Cr}}$ for Fe-ion-irradiated Eurofer 97 at 400 °C (figure), the size of the Cr clusters is $\sim 2.5 \text{ nm}$. Using the PCF model, the number density of clusters enriched as high as 20 at.% in Cr was evaluated at less than $4.2 \times 10^{23} \text{ m}^{-3}$.

5. Radiation-induced hardening

The contributions to hardening from different obstacle types within the framework of the dispersed barrier hardening (DBH) model [14] are in general considered by a superposition $\Delta\sigma_{\text{tot}} = \Delta\sigma_{\text{LR}} + \Delta\sigma_{\text{SR}}$ of long-range $\Delta\sigma_{\text{LR}}$ and short-range $\Delta\sigma_{\text{SR}}$ obstacles. The hardening caused by the interaction of moving dislocations with long-range stress field caused by dislocation network is $\Delta\sigma_{\text{disl}} = M_T \alpha_{\text{disl}} \mu b \sqrt{\rho_{\text{disl}}}$, where $\mu = 84.4 \text{ GPa}$ is the shear modulus of the matrix, $b = 0.248 \text{ nm}$ is the Burgers vector of gliding dislocations, α_{disl} is the dislocation barrier strength, $M_T = 3.06$ is the Taylor factor and

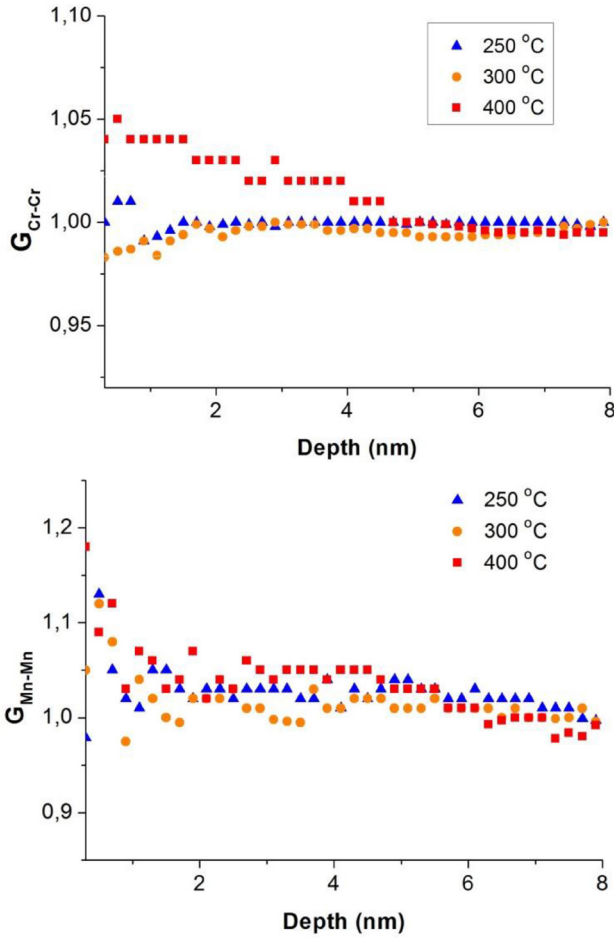


Figure 3. Cr–Cr and Mn–Mn correlation functions of Eurofer 97 irradiated with 5.6 MeV Fe²⁺ ions up to ~6 dpa.

ρ_{disl} is the dislocation density. The hardening from short-range obstacles of type i such as loops, voids, precipitates, etc, within the framework of the Orowan model is $\Delta\sigma_i = M_T \alpha_i \mu b \sqrt{N_i d_i}$, where N_i and d_i are the number density and the mean size of obstacles, respectively.

The contribution to hardening from dislocation loops of $\Delta\sigma_{\text{loops}}$ (see figure 4) was calculated within the framework of the DBH model with the value of the barrier strength of $\alpha_{\text{loops}} = 0.6$ [5, 15]. The hardening calculated with this value correlates with the increase of yield strength of Eurofer 97 after neutron irradiation at 300 °C (figure 5). This dislocation loop hardening for Eurofer 97 irradiated by ion beam to 10 dpa corresponds to one calculated by the DBH model with renewed dislocation loop data [6] for Eurofer 97 irradiated by neutrons to 15 and to 32 dpa.

The comparison of neutron and ion irradiation experimental results involves a lot of effects that should be taken into consideration [16]. We have compared the hardening calculated with the DBH model, which takes into account radiation-induced dislocation loops (see figure 4), with the change of yield stress for neutron-irradiated Eurofer 97 acquired in the SPICE irradiation project in the range of 250 °C–400 °C [17, 18]. The temperature shift calculated according to Mansur [19] was used to take into account the dose rate effect in the ion irradiation experiment.

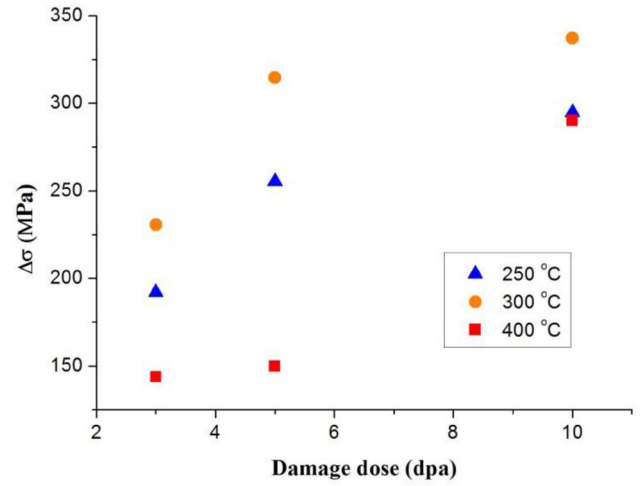


Figure 4. Dislocation loop hardening calculated with the DBH model versus the damage dose for Eurofer 97 irradiated with 5.6 MeV Fe²⁺ ions to $1 \times 10^{20} \text{ m}^{-2}$ at different irradiation temperatures.

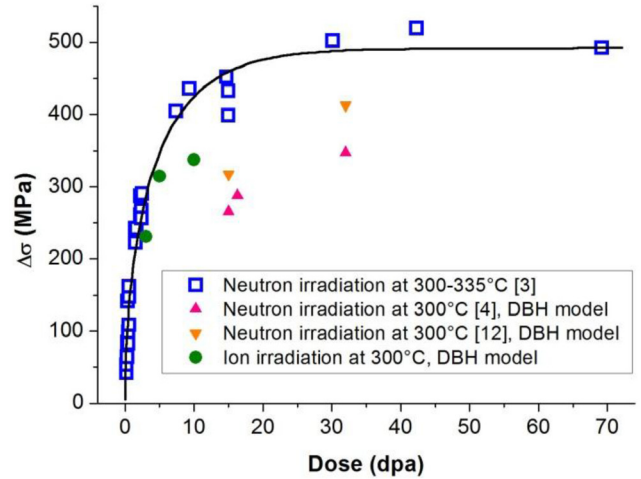


Figure 5. Change of yield stress versus irradiation dose for Eurofer 97. Open symbols (□) represent data for neutron-irradiated steel [3]; full symbols represent the result of calculation with the DBH model for dislocation loop hardening for neutron-irradiated steel (▲ [4], ▼ [6]) and for the steel irradiated with 5.6 MeV Fe²⁺ ions (●).

The comparison of hardening values obtained in the ion and neutron irradiation experiments (see figure 6) has shown a reasonable agreement.

It seems that the formation of dislocation loops plays the main role in the low-temperature radiation hardening of Eurofer 97 at the dose up to ~10 dpa. The hardening due to nucleation of Cr-enriched embryos calculated on the PCF model [11] and DBH model with the value of the barrier strength of $\alpha_{\text{Cr clust}} = 0.03$ [20] is less than 50 MPa for Eurofer 97 irradiated to 6 dpa at 400 °C and it is considerably smaller than the revealed dislocation loop hardening under the same irradiation conditions.

Slightly high hardening after Fe ion irradiation to 10 dpa at 400 °C is visible in figure 6. This effect can be the reason for

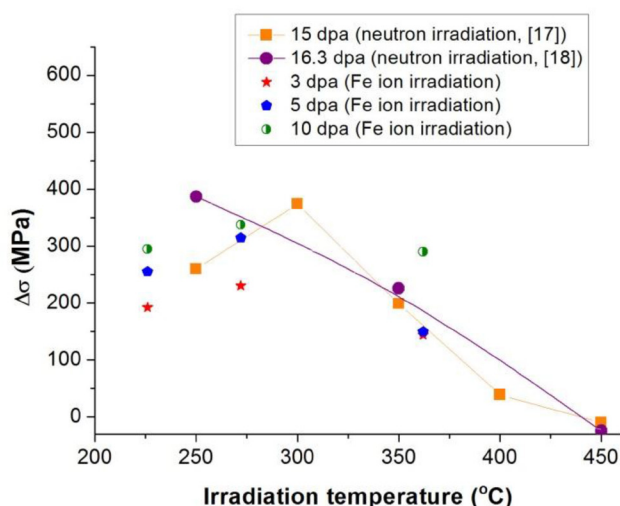


Figure 6. Change of yield stress versus irradiation temperature for Eurofer 97.

an increase of mean square deviation of loop sizes under these irradiation conditions. Another reason can be Cr-enriched embryos, as the largest of them can be detected as small dislocation loops. As a result, dislocation number density and their contribution to hardening were overestimated for this irradiation condition (see table 2 and figure 4).

The contribution to the radiation hardening can give voids. However, TEM analysis did not find visible voids in Fe-ion-irradiated Eurofer 97 samples. The estimated concentration of non-annihilated vacancies was less than 10^{-3} at.% and the contribution to the hardening from these vacancies was negligible. It should be mentioned that voids/helium bubbles nucleate due to helium creation from boron in neutron-irradiated Eurofer 97. For example, in Eurofer 97 irradiated to 16.3 dpa, detected cavities were distributed nonhomogeneously and were rare [4].

6. Results

In this work, the analysis of the microscopic origin of irradiation-induced hardening of Eurofer 97 under low-temperature irradiation with 5.6 MeV Fe^{2+} ions to $1 \times 10^{20} \text{ m}^{-2}$ at 250, 300 and 400 °C was carried out. The irradiated samples were characterized by TEM and APT. A preferable formation of dislocation loops was detected after the irradiation at all the temperatures used in the study. The pair-correlation analysis detected an initial stage of solid solution decomposition in Eurofer 97 irradiated to ~10 dpa. The detected microstructural changes and hardening calculated with the DBH model conform to the results for the neutron irradiated Eurofer 97.

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